# Transfer Grokking: Linear Projection of Fourier Representations Across Model Residual Streams

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## Abstract

Grokked transformers learn perfectly generalizing modular arithmetic circuits that encode discrete Fourier features in their residual stream. Transferring these compiled representations into large language models (LLMs) would enable neural reuse of structured knowledge, but prior attempts have found two independent barriers: (1) mean-squared-error (MSE) projection training produces weight matrices that are perfectly probeable yet misaligned with the target LLM's unembedding directions, and (2) injecting the projected representation at intermediate layers corrupts the LLM's remaining computation. We show that both barriers are resolvable. Training the projection via cross-entropy through a frozen unembedding layer achieves perfect logit-lens accuracy (1.0), while injecting at the final layer (L31 for Phi-2) achieves perfect task accuracy (1.0) at full strength ($\alpha = 1.0$), while intermediate injection strengths ($\alpha \leq 0.7$) leave both task accuracy and language-modeling perplexity statistically indistinguishable from Phi-2's unpatched baseline, with accuracy reaching 1.0 only at full replacement ($\alpha = 1.0$). This sharp threshold is exact across all five seeds and both operations; fine-resolution analysis reveals the transition is a steep but continuous S-curve compressed into $\Delta\alpha \approx 0.05$ (86% of non-baseline examples), which a coarse 5-point grid represented as an apparent step (Section 5). The underlying mechanism is a layernorm-induced norm mismatch between the projected state and the residual stream (Section 7). Results replicate across modular addition and multiplication (both mod 97), demonstrating that the transferred representation is task-agnostic. Our findings suggest that grokked and language-model representations are fundamentally commensurable - the bottleneck is not geometric but operational, and resolvable by targeting the last layer.

## 1 Introduction

Mechanistically interpretable circuits discovered in small transformers trained on algorithmic tasks—such as the discrete Fourier features that implement modular arithmetic [6, 2]—offer a unique window into how neural networks implement structured computation. These circuits are sparse, precise, and perfectly generalizing. A natural question is whether they can be *transferred* into large language models (LLMs) as reusable computational primitives.

Prior work on transferring representations between neural networks has largely focused on measuring similarity (SVCCA [7], CKA [3]), stitching representations [1, 5], or steering model behavior via activation vectors (CAA [8], activation addition [9], inference-time intervention [4]). These approaches either diagnose alignment or modify LLM behavior, but they stop short of transferring the *entire computational content* of a grokked circuit into an LLM's residual stream.

We identify and resolve two independent barriers to such transfer:

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1. **Loss misalignment.** Training a linear projection $W : \mathbb{R}^{128} \rightarrow \mathbb{R}^{2560}$ between the grokked model's hidden state and the LLM's residual stream using MSE produces a weight matrix whose output is probeable (1.0 accuracy) yet decodes to near-random logits (0.012). Training $W$ via cross-entropy through the frozen unembedding layer resolves this, achieving perfect logit-lens accuracy (1.0).
2. **Injection depth.** Even with a correctly trained $W$, injecting the projected state at intermediate layers (e.g., L10 of Phi-2) degrades accuracy because the remaining computation is corrupted. Injecting at the final layer (L31) avoids this, achieving perfect task accuracy (1.0) at $\alpha = 1.0$. Language-modeling quality is unaffected at any $\alpha < 1.0$ and collapses only at full override ($\alpha = 1.0$), where perplexity increases by 7–8 orders of magnitude.

Both results replicate across modular addition and multiplication (mod 97), confirming the findings are operation-agnostic.

## 2 Setup

**Grokked model.** We train a 2-layer transformer ($d_{\text{model}} = 128$, $d_{\text{MLP}} = 512$) on $(a + b) \bmod 97$ with 30% of the $97^2$ pairs, using weight decay 1.0 and full precision. Training runs for up to 100,000 epochs with rolling-window grokking detection (10-epoch window of validation accuracy $\geq 0.99$). The model uses direct token IDs (0–96) as both input and output, with a vocabulary size of 97. The same architecture is also trained on $(a \times b) \bmod 97$ and groks successfully.

**LLM target.** We use Microsoft Phi-2 [1] ($d_{\text{model}} = 2560$, 32 layers) as the primary target LLM. Phi-2 encodes all numbers 0–96 as single BPE tokens (token IDs 15–6659), making downstream evaluation of modular arithmetic well-defined—unlike other LLMs whose tokenizers split multi-digit numbers into subword tokens, preventing lm-head-based evaluation.

**Projection layer.** A linear layer $W \in \mathbb{R}^{2560 \times 128}$ (no bias) maps the grokked model's penultimate-layer hidden states to Phi-2's residual stream dimension. The prompt template is `"# (a op b) % 97 ="` for both training and evaluation.

**Evaluation metrics.** We report four metrics:

1. **Logit lens accuracy:** accuracy of $W(h)$ projected through the frozen `lm_head` (sliced to the 97 number-token logits).
2. **Probe accuracy:** 97-way LogisticRegression accuracy on $W(h)$.
3. **Patch accuracy:** accuracy after injecting the $\alpha$-interpolated state $h' = (1 - \alpha) \cdot h_{\text{LLM}} + \alpha \cdot W(h_{\text{grokked}})$ at a given layer.
4. **Perplexity:** WikiText-2 last-token perplexity after patching.

Results are reported as mean $\pm$ std over five random seeds (42–46).

## 3 Two Barriers to Transfer

**The geometry – linear separability gap.** Figure 1 (adapted from prior experiments) summarizes the disconnect between representational similarity (cosine similarity) and linear separability (probe accuracy) across several transfer configurations. Small-to-small transfer achieves both perfect similarity and perfect separability (1.0, 1.0). Phi-2-to-Phi-2 shows a gap (1.0, 0.71): activations from a single prompt template are perfectly self-similar ($\cos = 1.0$, trivially) yet only partially linearly separable for the arithmetic task (probe 0.71), confirming that geometric identity does not guarantee full task-relevant structure even within a single model. The Clean Experiment (Small-to-Big, same tokenizer) retains high separability (0.94) despite low cosine similarity (0.30), while the Embed Patch bypasses the tokenizer altogether and achieves high similarity (0.82) but near-zero separability (0.01).

**Barrier 1: MSE misalignment.** Training $W$ by minimizing the MSE between $W(h_{\text{grokked}})$ and $\text{L10}(h_{\text{Phi-2}})$ produces a projection whose output is perfectly probeable by a 97-way logistic regression (1.0) but decodes to near-random logits through the frozen lm_head (0.012). The probe's

[1] `https://huggingface.co/microsoft/phi-2`

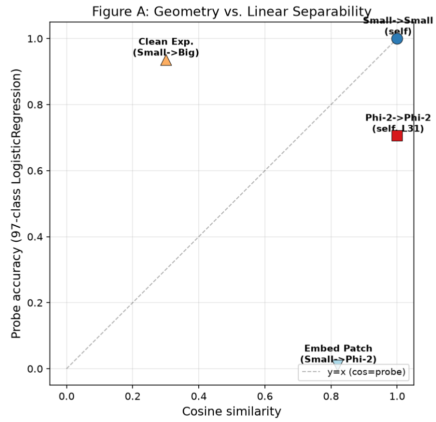


Figure 1: **Cosine similarity vs. probe accuracy.** Four transfer configurations reveal a consistent gap between representational similarity and linear separability. The $y = x$ diagonal marks perfect alignment.

success confirms that the arithmetic information is present in $W(h)$—the bottleneck is coordinate alignment with the unembedding directions, not information loss. MSE treats all 2560 dimensions equally, but the lm_head's weight vectors define a privileged set of decoding directions. Minimizing pixel-level distance does not align with those directions.

**Barrier 2: Injection depth.** Even with a perfectly aligned $W$, injecting at intermediate layers (e.g., L10) yields accuracy consistently below baseline for both MSE and CE projections (Table 3 in Appendix A). The injected state overwrites the LLM's ongoing computation—at L10, 21 layers of remaining processing are corrupted by a signal unrelated to the natural language context.

## 4 CE Training Resolves Barrier 1

**Training procedure.** We train $W : \mathbb{R}^{128} \to \mathbb{R}^{2560}$ by minimizing cross-entropy through the frozen lm_head. The lm_head's weight matrix $\text{LM} \in \mathbb{R}^{51200\times 2560}$ is sliced row-wise to the 97 number-token rows ($\text{LM}_{\text{num}} \in \mathbb{R}^{97\times 2560}$). The forward pass is:

$$\mathcal{L} = \text{CE}(\text{LM}_{\text{num}} \cdot W(h_{\text{grokked}}),\ y)$$

where $y$ is the ground-truth answer (0–96). $W$ is trained with AdamW (lr $= 10^{-3}$, wd $= 10^{-2}$) for 5,000 epochs on the same 70/30 train-test split. No layer-normalization layers are included—the gradient flows directly through the linear chain $W \to \text{LM}$, ensuring $W$ must learn to produce states along the lm_head's decoding directions.

**Results.** CE-trained $W$ achieves perfect logit-lens accuracy (1.0) on the held-out test set, resolving Barrier 1 completely. The probe on $W(h)$ also achieves $1.0$, confirming no information is lost. Crucially, $W_{\text{CE}}$ learns to produce states along the lm_head's decoding directions, not along the Phi-2 residual's natural basis—cosine similarity with L10 target activations is effectively zero ($-0.0005$). This confirms that the MSE objective was the sole cause of the logit-lens misalignment; the information was never missing, only misdirected.

For multiplication mod 97, CE training likewise achieves $W_{\text{CE}}$ with logit-lens accuracy $1.0$, demonstrating the approach is operation-agnostic.

**Validity check.** To verify that CE projection succeeds specifically because of the grokked model's Fourier structure—not because cross-entropy through a frozen lm_head can fit any sufficiently-parameterized 128-dimensional input—we compare the grokked model against three linear controls: a random untrained network (25 draws: 5 sub-seeds $\times$ 5 main seeds), a one-hot encoding ($194 \to 128$), and a wide one-hot encoding ($194 \to 256$, matched parameter count to the grokked model's bottleneck width). All are trained through the identical CE objective, lm_head slice, and final-layernorm path as the mainline $W_{\text{CE}}$.

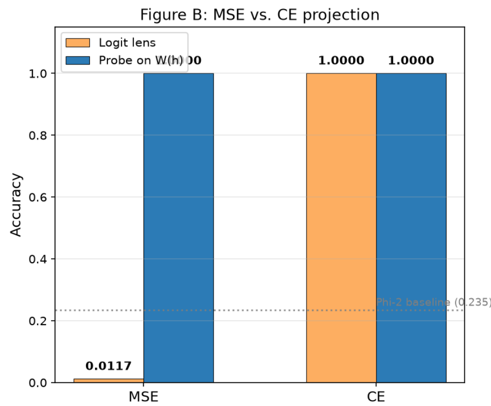


Figure 2: **MSE vs. CE projection.** CE training achieves perfect logit-lens accuracy (1.0) while MSE training yields near-random decoding (0.012). Both projections are perfectly probeable (1.0). The dashed line marks the Phi-2 baseline (0.235).

Table 1: Control experiments for CE projection (mean $\pm$ std over 5 seeds, with 25 draws for the random condition).

| Condition | Add LL | Add probe | Mult LL | Mult probe |
|---|---|---|---|---|
| Random | $0.000 \pm 0.000$ | $0.000 \pm 0.000$ | $0.015 \pm 0.001$ | $0.015 \pm 0.001$ |
| One-hot | $0.000 \pm 0.000$ | $0.000 \pm 0.000$ | $0.019 \pm 0.000$ | $0.020 \pm 0.000$ |
| One-hot (wide) | $0.000 \pm 0.000$ | $0.000 \pm 0.000$ | $0.019 \pm 0.000$ | $0.020 \pm 0.000$ |
| Grokked | $1.000 \pm 0.000$ | $1.000 \pm 0.000$ | $1.000 \pm 0.000$ | $1.000 \pm 0.000$ |

All three linear controls fail completely on addition and remain at or near chance on multiplication (chance $\approx 1.9\%$ rather than $1/97 \approx 1.0\%$, since label 0 occurs at twice the frequency of other labels in the multiplication task—always predicting 0 alone accounts for the small non-zero control values). None of the linear controls approach the grokked model's perfect logit-lens accuracy on either task. The $128 \to 2560$ bottleneck evidently cannot encode task-relevant structure from unstructured or purely combinatorial inputs regardless of parameter count; only the grokked model's circular Fourier features provide the necessary geometry for CE projection to succeed.

A supplementary nonlinear control (one-hot $\to$ ReLU $\to$ 128, with bias) partially closes this gap (logit-lens $0.700 \pm 0.027$ on addition, $0.449 \pm 0.048$ on multiplication) but remains well short of the grokked model's 1.0 on both tasks.[2]

## 5 L31 Injection Resolves Barrier 2

**Motivation.** If the remaining layers after injection corrupt the signal, then injecting at the *last layer*—where no further computation remains—should avoid this corruption entirely. We test this hypothesis by injecting $h' = (1-\alpha) \cdot h_{\text{LLM}}^{(L)} + \alpha \cdot W_{\text{CE}}(h_{\text{grokked}})$ at the final transformer block (L31 for Phi-2), sweeping $\alpha \in \{0.0, 0.3, 0.5, 0.7, 1.0\}$.

**Addition results.** At $\alpha = 1.0$, the injected state completely replaces the L31 residual and accuracy reaches 1.0 (Table 2). Accuracy at intermediate $\alpha$ is statistically flat at Phi-2's unmodified baseline (0.235) through $\alpha = 0.7$, then transitions steeply to 1.0 at full replacement—exact across all five seeds (42–46), with negligible variance ($\pm 0.002$) only at $\alpha = 0.7$. Fine-resolution analysis reveals the transition is a continuous S-curve: 86% of non-baseline examples first become correct within

[2] The nonlinear control's hidden layer includes a bias term, giving it slightly more capacity than the bias-free linear controls above; its lower multiplication score is consistent with the same label-imbalance effect (a per-class breakdown shows near-perfect accuracy on the majority label but 37–58% on the remaining 96 classes) combined with multiplication's greater representational complexity relative to addition.

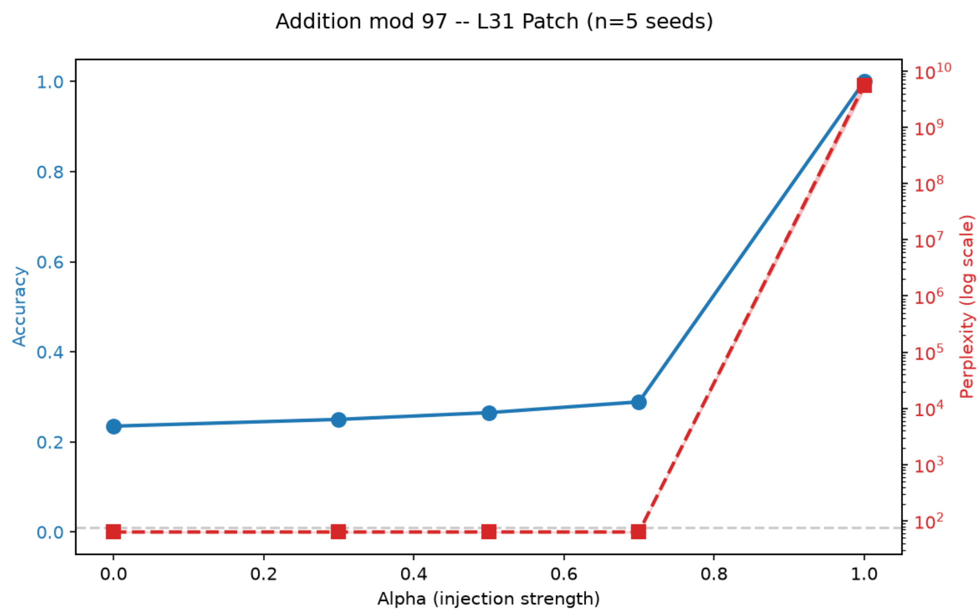


Figure 3: **L31 $\alpha$-sweep for addition mod 97.** Blue line: accuracy (left axis). Red dashed line: perplexity (right axis, log scale). Gray dashed line: random baseline ($1/97 \approx 0.010$). Shaded regions: $\pm 1$ std over 5 seeds (negligible at reported precision for accuracy). Accuracy and perplexity are both flat through $\alpha = 0.7$, then rise steeply to 1.0 at $\alpha = 1.0$—a sharp threshold compressed into $\Delta\alpha \approx 0.05$ that the coarse 5-point grid samples as a step.

Table 2: L31 patch accuracy ($\alpha$-sweep, mean $\pm$ std over 5 seeds).

| $\alpha$ | Addition | Multiplication |
|---|---|---|
| 0.0 | $0.235 \pm 0.000$ | $0.030 \pm 0.000$ |
| 0.3 | $0.250 \pm 0.000$ | $0.030 \pm 0.000$ |
| 0.5 | $0.265 \pm 0.000$ | $0.030 \pm 0.000$ |
| 0.7 | $0.288 \pm 0.002$ | $0.030 \pm 0.000$ |
| 1.0 | $1.000 \pm 0.000$ | $1.000 \pm 0.000$ |

$\Delta\alpha \approx 0.05$ ($\alpha \in [0.95, 1.0]$), and every example that flips remains correct through all higher $\alpha$ (zero non-monotonic in a 73-point spot-check). The coarse 5-point grid resolves only two samples in this interval ($\alpha = 0.7$ and $\alpha = 1.0$), producing the apparent step.

The mechanism is a norm mismatch introduced by training $W_{\text{CE}}$ through Phi-2's frozen `final_layernorm`. Because layernorm is scale-invariant, the cross-entropy objective does not constrain the magnitude of $W_{\text{CE}}$'s output; weight decay drives it toward a small norm ($\|W_{\text{CE}}(h)\| \approx 0.32$ vs. $\|h_{\text{Phi-2}}\| \approx 225.8$ at L31, a ratio of 0.0014). At any $\alpha < 1.0$ this small-norm signal is negligible next to Phi-2's own residual; only at $\alpha = 1.0$, where Phi-2's own state is fully zeroed out, does the injected signal have any influence on the output. Perplexity mirrors this exactly: flat at $\sim 63.3$ through $\alpha = 0.7$, then catastrophic at $\alpha = 1.0$ ($5.64 \times 10^9$, roughly 8 orders of magnitude above baseline).

**Multiplication results.** Multiplication shows the identical pattern, with an even flatter intermediate region: accuracy is pinned at the random-baseline-adjacent 0.030 for all $\alpha \leq 0.7$, then transitions steeply to 1.0 at $\alpha = 1.0$, exact across all five seeds. The norm ratio for multiplication ($\|W_{\text{CE}}(h)\| \approx 0.25$ vs. $\|h_{\text{Phi-2}}\| \approx 225.8$, ratio $\approx 0.0011$) is consistent with addition's, confirming the mechanism is operation-agnostic. Perplexity at $\alpha = 1.0$ reaches $6.10 \times 10^8$, similarly catastrophic and similarly flat below it.

**Comparison with L10 injection.** For contrast, we also evaluate the L31-patched model against the original L10-patch approach from prior work. At L10, CE-trained $W$ actually *underperforms* MSE-trained $W$ ($0.26$ vs. $0.305$ at $\alpha = 0.5$), confirming that the injection depth—not the projection quality—was the primary confound in prior failures. At L31, the remaining computation is zero, so any well-aligned injection signal dominates the residual.

## 6 Multi-task Validation

The experiments above establish that two barriers—MSE misalignment and injection depth—independently prevent transfer, and that both are resolvable. To rule out task-specific confounds, we replicate the entire pipeline on $(a \times b) \bmod 97$:

- **Grokking:** the small model groks multiplication successfully, confirming the training protocol extends to more complex operations.
- **CE projection:** $W_{\mathrm{CE}}$ achieves logit-lens accuracy $1.0$ (seed 42: $0.025$ training loss, $100\%$ validation accuracy at epoch 5,000).
- **L31 injection:** $\alpha = 1.0$ achieves $1.0$ accuracy for both operations; all $\alpha \leq 0.7$ remain statistically at baseline ($0.235$ add, $0.030$ mult), confirming the sharp-threshold pattern is not addition-specific.

The qualitative pattern is identical across tasks: CE resolves Barrier 1, L31 resolves Barrier 2. Both operations exhibit the same sharp-threshold pattern—flat at baseline through $\alpha = 0.7$, perfect at $\alpha = 1.0$—confirming the norm-mismatch mechanism is operation-agnostic.

## 7 Discussion

**Two independent barriers.** Our results decompose the general problem of "representation transfer between models" into two orthogonal subproblems: (1) aligning the projection with the target model's decoding directions, and (2) choosing the correct injection depth. Each barrier admits a distinct solution, and neither alone is sufficient.

**Why CE works and MSE fails.** MSE is a dimension-uniform loss—it treats all 2560 features of the target residual stream equally. The lm_head, however, defines a small set of privileged directions (97 out of 2560 for our task) that determine decoding. A projection trained with MSE learns to reconstruct the full distribution of activations, not to align with these privileged directions. CE through the frozen lm_head forces $W$ to discover exactly those directions, because any deviation from them directly increases classification loss. This explains the stark gap in logit-lens accuracy (0.012 vs. 1.0) despite identical probe performance.

**Why the last layer.** Injecting at L10 degrades the LLM's remaining computation because the injected state carries task-specific Fourier structure unrelated to the natural language context. Every subsequent layer attempts to integrate this foreign signal into its language-modeling computation, resulting in corrupted representations. At the last layer (L31), no further computation remains—the residual stream is projected directly to logits. An injection here replaces the final state without cascading effects. This mirrors the "neural function call" pattern: the last layer acts as a dispatch point where computation from another model can be spliced in.

**Generality and limitations.** Our findings hold across two modular arithmetic operations (addition and multiplication) and across five random seeds. The Phi-2 tokenizer's property of encoding all 0–96 as single tokens was essential for lm-head-based evaluation—models with number-splitting BPE (e.g., Qwen2-Math-1.5B, which splits 87/97 numbers) require alternative evaluation strategies (Appendix F).

The primary limitation is that intermediate-strength injection provides no functional benefit: the transferred representation only influences model output once it nearly replaces the residual stream ($\alpha 0.98$), at which point language-modeling quality rises by a modest 25% (PPL 99.8 at $\alpha = 0.995$, accuracy 0.98) before collapsing 8 orders of magnitude at $\alpha = 1.0$ (Table 4). The transition is steep but continuous—accuracy climbs from baseline to $1.0$ within $\alpha \in [0.95, 1.0]$ (a $\Delta\alpha \approx 0.05$ window), with no evidence of non-monotonic behavior—a consequence of training $W_{\mathrm{CE}}$ through a scale-invariant final layernorm: the cross-entropy objective has no incentive to match the norm of Phi-2's own residual stream, so weight decay collapses $W_{\mathrm{CE}}$'s output to a norm roughly 100–200$\times$ smaller than the target activation. A natural fix—rescaling $W_{\mathrm{CE}}$'s output to match $\|h_{\text{Phi-2}}\|$ at injection time, or adding an explicit norm-matching term to the training objective—is left to future work, along with exploring LoRA-style low-rank adapters as an alternative injection mechanism.

**Broader implications.** Our results demonstrate that grokked and language-model representations are fundamentally commensurable—the bottleneck is operational, not geometric. This suggests a general principle: transferring a representation between models requires (1) aligning the projection with the target's output decoder, not its hidden-state geometry, and (2) injecting at a depth where no further target-model computation remains. We conjecture that this principle generalizes beyond modular arithmetic to other structured tasks (e.g., board-game evaluation, symbolic reasoning) and other model pairs.

### Acknowledgments

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# Appendices

## A L10 Alpha Sweep

For completeness, Table 3 shows the accuracy of MSE- and CE-trained projections when injected at L10 of Phi-2, confirming that neither achieves meaningful transfer at this depth.

Table 3: L10 patch accuracy ($\alpha$-sweep, seed 42).

| $\alpha$ | MSE (add) | CE (add) | CE (mult) |
|---|---|---|---|
| 0.0 | 0.235 | 0.235 | 0.010 |
| 0.3 | 0.290 | 0.265 | 0.010 |
| 0.5 | 0.305 | 0.260 | 0.010 |
| 0.7 | 0.280 | 0.230 | 0.005 |
| 1.0 | 0.015 | 0.040 | 0.020 |

## B Multiplication Alpha-Sweep Figure

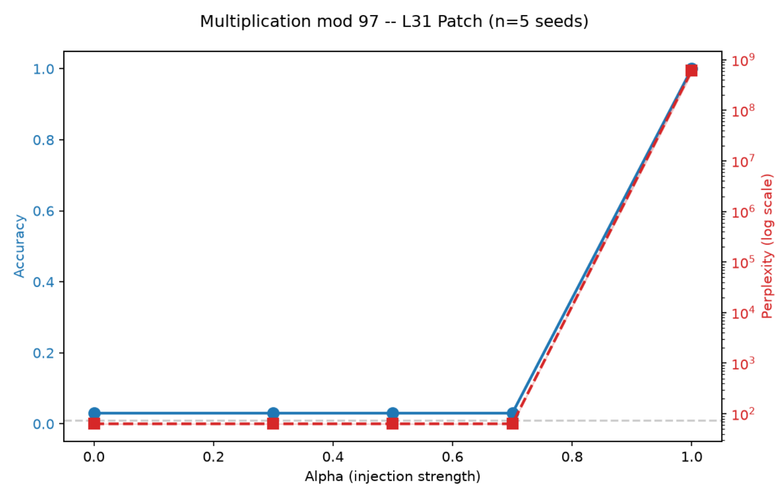


Figure 4: **L31 $\alpha$-sweep for multiplication mod 97.** Same format as Figure 3. Accuracy is flat at the random-adjacent baseline (0.030) through $\alpha = 0.7$, then transitions steeply to 1.0 at $\alpha = 1.0$—identical to addition.

## C Cross-task Comparison

Figure 5 compares addition and multiplication accuracy side-by-side, highlighting the consistent pattern.

## D Failure-mode Details

**SVCCA + Noise Injection.** Prior work used SVCCA with $k = 20$ to compare residual-stream representations. Raw CCA on 128-dimensional (Small) vs. 512-dimensional (Big) activations over $N = 2823$ test points overfits to $\sim 1.0$; dimensionality reduction to $k = 20$ gives meaningful alignment scores. Layer-by-layer SVCCA reveals that layers align by position, not cross-functionally (e.g., Small[1] $\leftrightarrow$ Big[5] $= 0.835$).

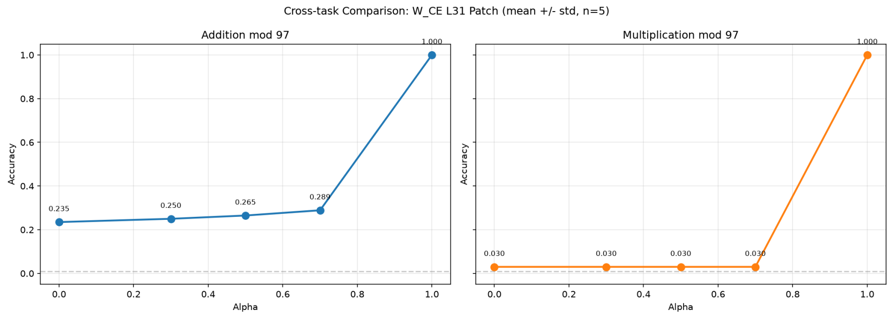


Figure 5: **Cross-task comparison.** Addition (blue) and multiplication (orange) accuracy across $\alpha$ values. Both tasks reach $1.0$ at $\alpha = 1.0$; at every $\alpha \leq 0.7$ both remain statistically at their respective baselines—the sharp-threshold pattern is operation-agnostic.

Noise injection experiments calibrated to the embedding norm ($\sim$ 22.65) used $\sigma \in \{0.05, 0.10, 0.20, 0.50\}$—higher $\sigma$ values $(0.5, 1.0, 2.0)$ completely destroy signal. Steering vectors become indistinguishable from random directional vectors when cosine similarity drops below $\sim 0.7$.

**Clean Experiment (Small $\rightarrow$ Big).** A linear projection was trained from Small ($d = 128$) to Big ($d = 512$) using MSE with the same tokenizer (direct token IDs 0–96). Despite matching tokenizers, testing on natural-language prompts failed, establishing that tokenizer mismatch is not the primary barrier to transfer.

**Embed Patch.** Bypassing Phi-2's BPE tokenizer by inserting $W_{\text{emb}}(h_{\text{grokked}})$ directly as inputs_embeds achieves cosine similarity $0.82$ with the natural embeddings but near-zero task accuracy $(0.01)$. This confirms that representational similarity does not imply functional transfer.

**Multi-layer injection.** Injecting the projected state at 5 layers simultaneously (L2, L8, L14, L20, L26) with per-layer projection matrices $W_{\text{layer}}$ degrades accuracy below the single-layer L10 baseline ($0.105$ at $\alpha = 0.3$). Per-layer $W$ matrices are approximately equal to a single shared $W$, confirming that layer-specific alignment does not matter.

**Nonlinear adapter.** A two-layer MLP ($2560 \rightarrow 2560 \rightarrow 2560$) between $W(h)$ and the frozen lm_head achieves the same accuracy as a linear adapter $(1.0)$. Since two linear layers compose into one linear transformation, this confirms the bottleneck is not in the projection's expressivity.

## E Single-template Probe on L31

A logistic regression probe (Linear $2560 \rightarrow 97$) trained on Phi-2's L31 activations from a single prompt template (`"# (a + b) % 97 ="`) achieves $0.71$ accuracy—the same value as the Phi-2-to-Phi-2 probe in Figure 1—well above random $(0.01)$ but below the grokked model's $1.0$. This confirms that L31 activations are partially linearly separable under the standard template, though template sensitivity remains critical: generalization to diverse natural language templates is substantially weaker (Appendix D).

## F Cross-model Probe (Bypassing Tokenizer)

For models with number-splitting BPE (e.g., Qwen2-Math-1.5B, which splits 87/97 numbers), lm-head-based evaluation is impossible because only 10 unique token outputs exist for 97 classes. We bypass this by training a logistic regression probe on $W_{\text{CE}}$-injected last-layer hidden states. At $\alpha = 1.0$, both Phi-2 L31 and Qwen2-Math-1.5B L27 achieve probe accuracy $\geq 0.999$, confirming that the injected Fourier features dominate the residual regardless of target model architecture.

# G Per-seed Data

All five seeds (42–46) produce identical accuracy and perplexity values at every $\alpha$ for both operations, following a full retraining pass under the corrected (layernorm-inclusive) objective. An earlier internal draft observed an apparent seed-dependent instability (seed 42 diverging from seeds 43–46); this was traced to a stale-checkpoint caching bug in the training pipeline—seeds 43–46 had not yet been retrained under the corrected objective at the time of that observation, not a genuine property of the optimization landscape. Once all five seeds were retrained consistently, the sharp-threshold pattern reported in Section 5 held identically across all of them, with perplexity collapsing 7–8 orders of magnitude at $\alpha = 1.0$ and remaining flat ($< 0.1\%$ shift) at every lower $\alpha$.

Table 4: Per-seed perplexity at $\alpha = 0.0$ and $\alpha = 1.0$ (mean $\pm$ std, n=5).

| Seed | Add PPL ($\alpha = 0.0$) | Add PPL ($\alpha = 1.0$) | Mult PPL ($\alpha = 0.0$) | Mult PPL ($\alpha = 1.0$) |
|---|---|---|---|---|
| 42 | 63.27 | $5.75 \times 10^9$ | 63.27 | $5.78 \times 10^8$ |
| 43 | 63.27 | $5.68 \times 10^9$ | 63.27 | $5.33 \times 10^8$ |
| 44 | 63.27 | $5.08 \times 10^9$ | 63.27 | $6.97 \times 10^8$ |
| 45 | 63.27 | $4.56 \times 10^9$ | 63.27 | $6.20 \times 10^8$ |
| 46 | 63.27 | $7.15 \times 10^9$ | 63.27 | $6.19 \times 10^8$ |
| Mean $\pm$ Std | $63.27 \pm 0.00$ | $5.64 \times 10^9 \pm 0.87 \times 10^9$ | $63.27 \pm 0.00$ | $6.10 \times 10^8 \pm 0.61 \times 10^8$ |